

MD-SIG Standards Released for Lighting LED Driver Interface

Specifications for a universal electrical interface between LED lighting modules and their power supplies have been recently released by Module-Driver Interface Special Interest Group (MD-SIG). This standard is supported by industry players like Osram, Panasonic, Philips and Tridonic.

According to the press release, "The standard is called LEDset1 edition1.0 (a 3-wire interface) power and Iset, and a return for both. The specification also allows for separate returns for mains and set currents to avoid setting errors due to return voltage drops."

Adding further, the release states that Iset has nothing to do with dimming. It is provided so that the LED module can indicate its maximum working current to the driver. It does this by drawing $I_{max}/1000$ from Iset. The only time any modulation of Iset is allowed is if the LED module is overheating, when it is permitted to reduce its own supply current by reducing Iset.

Official names of the three terminals are LED+, LEDset and LED, as stated in the release. The optional Iset return is GRNset. Additionally, the LED module and driver must be marked accordingly. The recommended wire colours are red, white and black, respectively. Although LEDset1 specifies a universal analogue signalling protocol, it does not cover the matching of the driver output voltage and current capability with LED module's needs.

"The power interface specification describes driving capabilities like voltage, current and power ranges of LED control gears and respective LED module operating requirements," says the SIG. "It simplifies matching and comparing output and input parameters with harmonised terms and definitions."

Source: ChipsnWafers.com

Quantum dots has also brought out a new LED driver named SEFA which functions in three modes of operation, namely, BCM, CCM and DCM. Babu explains that there is no complexity in the design part and it does not suffer from high voltage stress in the power switch.

Infineon has also brought out ICL8105 single-stage fly-back controller with an optional burst-mode control scheme that helps extend the dimming range. It is aligned to line frequency to avoid noise, flickering or shimmering.

JSK Innovative Technology Pvt Ltd has just completed their research and development of a phase-cut dimmable driver with a range starting from 15V to 55V, current range of 100mA to 700mA and above 0.95

power factor correction (PFC).

Fairchild has also announced the launch of their phase-cut dimmable single-stage LED driver integrated circuit (IC) called FL7733A. This is a highly-integrated pulse width modulator (PWM) controller that provides tight-tolerance constant-current output using TRUECURRENT technology. Fairchild claims that this LED driver can enable designs with constant current tolerance of less than ± 1 per cent over the universal line voltage range.

High intensity. Toshiba Semiconductor has launched TL114 series of high-power white LEDs that achieve luminous flux of 140lm (min) as compared to 130lm (min) previously.

Plessey has announced a range of MaGIC LED die manufactured on

GaN-on-Silicon. This process technology will form the base for their applications-specific LEDs.

Circuit protection. "We have seen many automation systems getting damaged due to the reverse current of drivers. So our new range of dimmable drivers have an opto-isolation circuit designed into it that protects the system from reverse current in the dimmable driver," says Rakesh Shah, managing director, JSK Innovative Technology Pvt Ltd (JSK Urja).

Littlefuse has launched their LSP05 and LSP10 thermally-protected surge protective devices designed for outdoor and commercial LED fixtures. These devices use thermally-protected varistor technology for transient over-voltage protection.

Better packaging

"Since last year, some changes that have taken place include the change from emitter technology to chip on board (CoB) technology and different soldering processes followed by various chip manufacturers," says Siva Satheesh, managing director, and Abdul Basith, executive director, Euro lights, India.

An active PFC greater than 0.95 is claimed to be attainable with the newly-launched RCOB LED drivers for CoB LEDs from RECOM. Active PFC diminishes total harmonics, automatically corrects for AC input voltage and is capable of delivering a full range of input voltage.

LED Engin claims that they have used an improved LSP package design and a low thermal resistance per package footprint of $0.5^{\circ}\text{C}/\text{W}$ to build a 25-die, 80W emitter that can deliver 3800lm from a

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